

# THE STORAGE AND I/O WALL IN EXASCALE COMPUTING

Modern GPU-accelerated CFD simulations routinely generate datasets ranging from hundreds of gigabytes to multiple terabytes per snapshot, making storage, I/O, and data movement major bottlenecks in exascale scientific workflows. As compute capability continues to scale, filesystem bandwidth and storage infrastructure increasingly limit output cadence, visualization, and downstream analysis.

## BOTTLENECKS



Multi-terabyte simulation snapshots



Filesystem and MPI-I/O bottlenecks



High storage and data transfer costs



Limited output cadence at extreme scale



Expensive post-hoc analysis workflows



Result: I/O and storage wall limits science and productivity

# OUR SOLUTION: SOLVER-INTEGRATED COMPRESSION

We integrate both **lossless compression** for restart/checkpoint reliability and **error-bounded lossy compression** for analysis-oriented outputs directly into the GPU-enabled spectral-element solver nekRS. By compressing solution fields *in situ* before MPI-I/O, we substantially reduce I/O volume and storage cost while preserving scientific value.

## BENEFITS & ENABLED WORKFLOWS



Reduces storage footprint and I/O volume



Minimizes data movement overhead



Preserves QoIs within controlled tolerances



Maintains scientific usability of reconstructed fields

## ENABLED WORKFLOWS



Scientific visualization



Post-hoc analysis



Reduced-order modeling (ROM)



Data-driven / ML workflows



Result: Lower I/O cost, higher output cadence, and scalable science

# METHODOLOGY: COMPRESSION-AWARE I/O FRAMEWORK

A portable, compression-aware I/O framework integrated in nekRS.



GPU Solver (nekRS)

- Spectral-element discretization
- High-order accuracy
- GPU-accelerated



Rank-local Fields (Element-based)

- Each MPI rank holds a subset of elements
- No inter-rank communication for compression



Compression (in situ)

- Error-bounded SZ3 for analysis
- SZ3 (lossy) / Blosc2 (lossless)
- Lossless Blosc2 for checkpoints
- Embarrassingly parallel



Portable Compressed Format (Self-describing)

- Rank-decomposable metadata
- Decouples storage from MPI layout
- Supports arbitrary read-time parallelism



Readers & Workflows (Coupled or Decoupled)

- Visualization • Post-hoc analysis
- ROM • ML / GNN pipelines
- In situ or ex situ access



QoI-aware Tolerance Selection

Tolerances chosen based on workflow needs: interpolation-dominated QoIs allow larger  $\epsilon$ ; derivative-sensitive QoIs require stricter  $\epsilon$ .



Solver-Integrated Pipeline

Compress solution fields *in situ* before MPI-I/O write to reduce data volume, I/O time, and storage cost.



Workflow Flexibility

Compressed datasets support visualization, analysis, ROM, and ML/GNN pipelines via coupled or decoupled readers.

## QoI PRESERVATION & TOLERANCE SELECTION

Compression accuracy is evaluated on physically relevant quantities of interest (QoIs), not just pointwise error.

### Analytical QoIs



Velocity magnitude  $|u|$



Kinetic energy  $E_k$



Turbulence kinetic energy (TKE)



Integral flow statistics

### Derivative-sensitive QoIs



Vorticity ( $\omega$ )



$\lambda_1$  structures



Gradients ( $\nabla u$ )



Wall-shear related quantities

### Fusion blanket dataset:

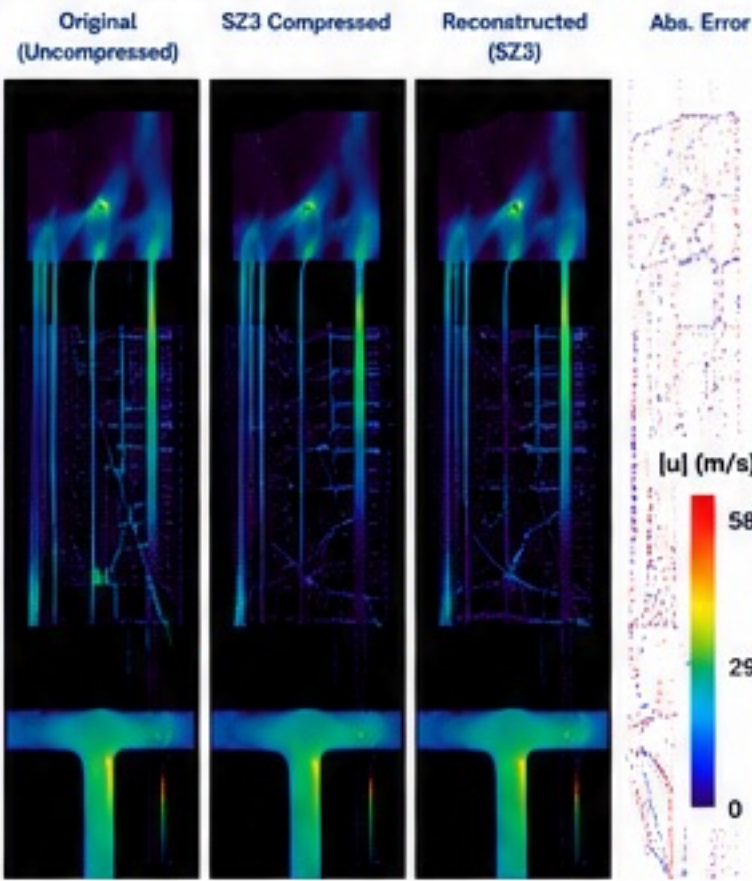
$$u_x \in [-58, 58] \text{ (m/s)} \quad \epsilon_{rel} = 10^{-3}$$



Observed velocity-magnitude error  $|u_{mag} - \hat{u}_{mag}| \leq 1.6 \times 10^{-3}$

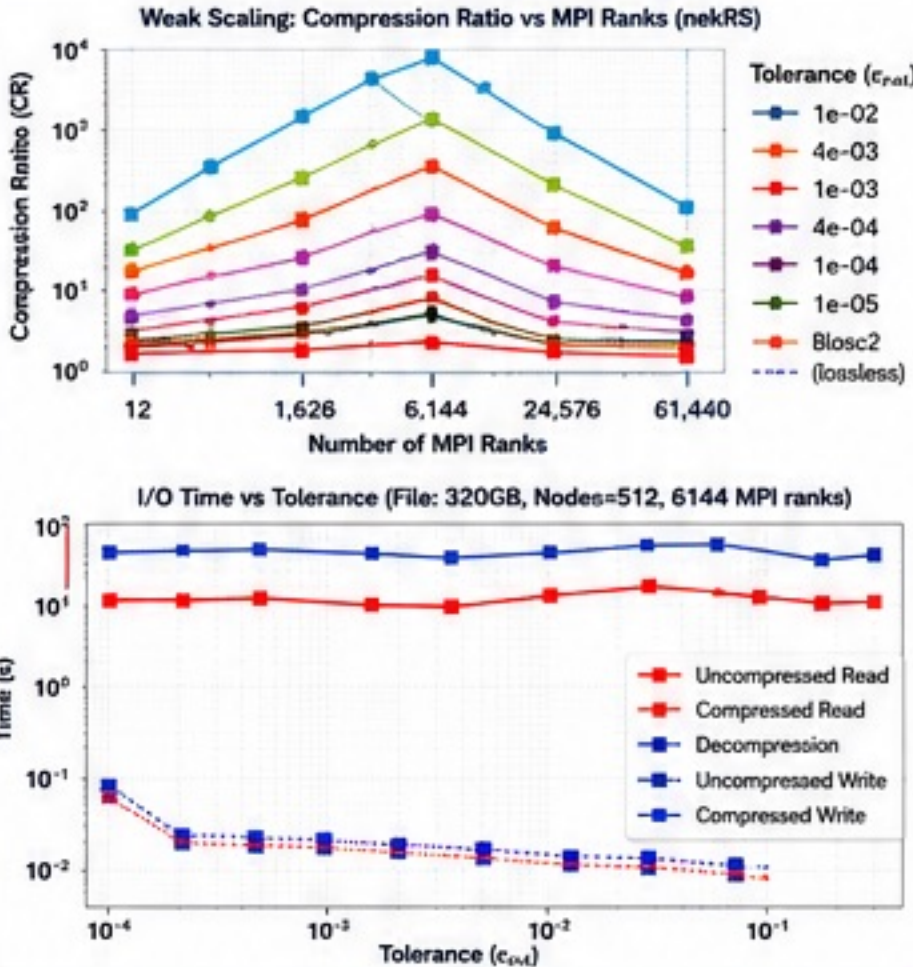
Key flow features and analytical QoIs are preserved within controlled tolerances.

## FUSION BLANKET: VELOCITY MAGNITUDE (SZ3 COMPRESSION, $\epsilon_{rel} = 10^{-3}$ )



Large-scale flow structures and features are well preserved after compression.

## PERFORMANCE: WEAK SCALING & I/O TIME



Up to 100x compression ratio and up to 8x I/O time reduction.

## GENERALIZATION



Applicable to a wide range of distributed scientific workflows with locally structured data layouts.



Extends to:

- High-order FEM / DG
- Structured-grid CFD
- Multiphysics workflows
- MPI-decomposed fields



Key ideas:

- MPI-rank-local compression
- In situ data reduction
- QoI-aware error-bounded compression
- Compressed parallel MPI-I/O

A general strategy for scalable, data-centric HPC workflows.

## IMPACT



Higher output cadence



Larger simulation campaigns



Improved filesystem scalability



Reduced storage & energy costs



Practical handling of multi-terabyte simulation datasets



Data reduction is as critical as numerical computation in the exascale era.

## FUTURE WORK



GPU-resident compression and asynchronous I/O



Tighter QoI-aware tolerance selection



Integration with visualization tools (ParaView / VisIt)



Support for *in situ* analysis and steering



SEM-aware transform-based compression strategies

### Additional directions:

- Reactive and multiphysics flows
- Adaptive tolerance selection
- Compressed ML-ready datasets for ROM and AI workflows

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TAKEAWAY: Compression ratio >100x and I/O time reduction up to 8x